# Kauaʻi – Coastal flooding ESF-6 playbook

**Data as of 2026-10-28**

**County:** Kauaʻi · **Hazard:** Coastal flooding

**Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.**

## 1. Summary

**Calculated by open code from public data · see Sources (section 8)**

| | | |
|---|---|---|
| People in the risk area<br>**19,186** | May need a public shelter (likely)<br>**959** | May need a public shelter (low – high)<br>**384 – 2,110** |
| Speak English less than well (C)<br>**487** | Use power-dependent medical equipment (estimate) (M)<br>**98** | People with a disability (I)<br>**1,915** |
| Age 65 and older (I)<br>**4,054** | People in households without a car (T)<br>**852** | Hearing difficulty (C)<br>**549** |
| Vision difficulty (C)<br>**408** | Cognitive difficulty (memory, decisions) (I)<br>**773** | Age 65 and older, living alone (S)<br>**863** |
| Known 2026 shelter sites usable for this hazard<br>**5** | | |

Source: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER, FEMA National Risk Index v1.20; shelter rates: section 8.

Share of people in the risk area who may need a public shelter: low 2%, likely 5%, high 11%. One person can be counted in more than one need. Page 2 explains how to read these numbers.

## 2. Scope: county total vs. one community

**The county total is an upper bound.** It counts the 25% of Census areas in the county with the highest FEMA risk score for this hazard as hit at the same time. Real events usually hit fewer communities.

## Largest single community (the Census areas with the most people who may need a shelter)

| Community (Census area) | People in the risk area | May need a public shelter (low – likely – high) | Speak English less than well | Use power-dependent medical equipment (estimate) | People with a disability | Age 65 and older | People in households without a car |
|---|---|---|---|---|---|---|---|
| Līhuʻe<br>Census area 405 | 7,301 | 146 – **365** – 803 | 276 | 33 | 856 | 1,388 | 120 |
| Kekaha / Waimea<br>Census area 409 | 5,172 | 103 – **259** – 569 | 140 | 26 | 410 | 1,077 | 365 |
| Kōloa / Poʻipū<br>Census area 406.03 | 2,972 | 59 – **149** – 327 | 49 | 19 | 284 | 801 | 254 |

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

To plan for one community, use the first row.

# How to read this playbook

## Example row

| Community (Census area) | People in the risk area | May need a public shelter (low – likely – high) |
|---|---|---|
| Līhuʻe<br>Census area 405 | 7,301 | 146 – **365** – 803 |

Read it as: about 7,301 people live in the risk area of Līhuʻe. If this hazard hits, between 146 and 803 of them may need a public shelter; 365 is the likely number.

## Census area

A neighborhood-sized area the U.S. Census uses, usually 1,200–8,000 people. The number (for example "Census area 308") is its Census ID. Each area is named after the Census place where most of its people live. "near" means no one in the area lives inside a named place; "rural near" means most people live outside named places.

## Low – likely – high

The share of people in the risk area who may go to a public shelter: low 2%, likely 5%, high 11%. Where the rates come from: Team assumptions: likely 5%; low 2% (the hurricane planning factor); high 11% (the Lahaina 2023 rate).

## Who has access and functional needs (CMIST)

- **C – Communication:** people who speak English less than well (they may need interpreters and translated messages), and people with hearing or vision difficulty.
- **M – Medical:** people who use power-dependent medical equipment (an estimate from HHS emPOWER).
- **I – Independence:** people with a disability, people with cognitive difficulty (memory, decisions) and people age 65 and older.
- **S – Safety and support:** people age 65 and older living alone, children and people living in group quarters (in the needs table).
- **T – Transportation:** people in households without a car.

## County total vs. one community

The county total counts the 25% of Census areas in the county with the highest FEMA risk score for this hazard as hit at the same time, so it is an upper bound. Use the "Largest single community" table to plan for an event that hits one community.

## Map

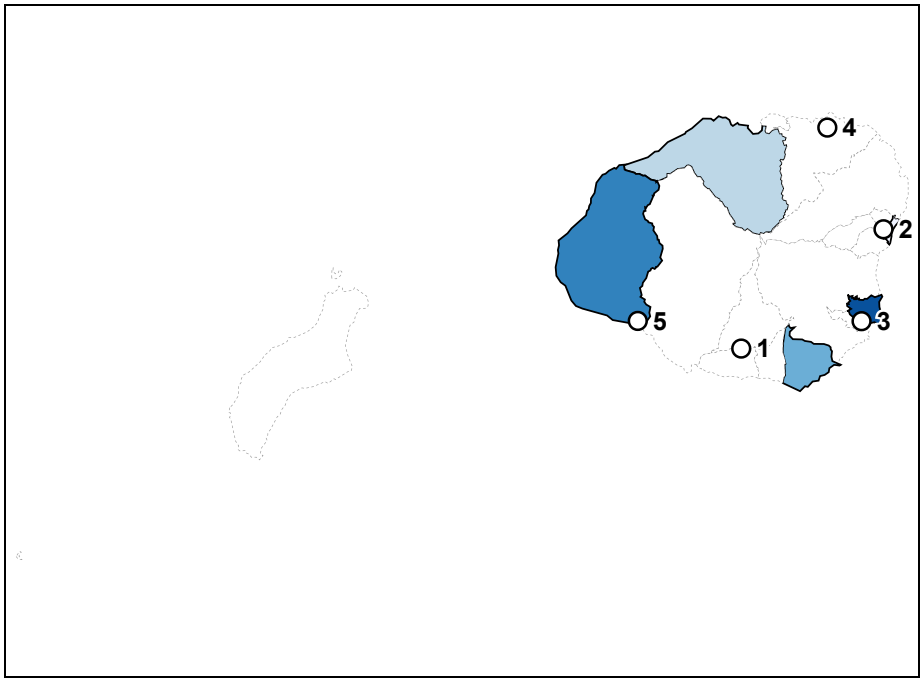
Darker blue means more people with needs, from "Lowest need" to "Highest need". A solid outline marks a Census area in the risk area; a dashed outline is outside it. Symbols: ○ usable site, △ check flood zone, ◇ flood risk not mapped, ⊠ not for this hazard. Numbers on the map match the site key.

## Why population numbers can differ

People in the risk area come from ACS 2018–2022 estimates (through the CDC Social Vulnerability Index 2022). Community names use 2020 Census counts. County totals in the plan use Census estimates for 2025. Each table names its source underneath.

**Every number is a planning estimate. In an event, follow county announcements.**

## 3. Map: people with needs and known sites




### Site key

**1** = Kalāheo Neighborhood Center (usable)

**2** = Kapa‘a Middle School (usable)

**3** = Kaua‘i High School Gym (usable)

**4** = Kīlauea Neighborhood Center (usable)

**5** = Waimea Canyon Middle School (usable)

- Lowest need (0–209 people)
- Lower (209–310 people)
- Middle (310–607 people)
- Higher (607–940 people)
- Highest need (940–1,285 people)
- Solid outline: Census area in the risk area
- Dashed outline: outside the risk area
- Usable site
- Check flood zone
- Flood risk not mapped (FEMA zone D)
- Not for this hazard

Color = people in the risk area who speak English less than well, use power-dependent medical equipment, have a disability, or live without a car (a person can be counted more than once). No street map: Census area outlines only.

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

**No 2026 activated site within a 15-minute drive (normal traffic)**, for sites usable for coastal flooding: 6 of 17 populated Census areas. With DOE candidate schools usable for this hazard: 5 of 17. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Drive-time matrix, per hazard: shortest drive from the area's land point to a 2026 activated site usable for that hazard on the same island; scenario 2 adds the 5 nearest DOE candidate schools usable for it. Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

# 4. Known sites (opened in 2026 events)

**Compiled from official county releases · verified by team**

| # | Status for this hazard | Site | Address | Why | Times opened in 2026 |
|---|---|---|---|---|---|
| 1 | ✓ **Usable** | Kalāheo Neighborhood Center | 4480 Papalina Road, Kalāheo | Outside FEMA high-risk flood zones (zone X) | 1 |
| 2 | ✓ **Usable** | Kapaʻa Middle School | 4867 Olohena Road, Kapaʻa | Outside FEMA high-risk flood zones (zone X) | 1 |
| 3 | ✓ **Usable** | Kauaʻi High School Gym | 3577 Lala Road, Līhuʻe | Outside FEMA high-risk flood zones (zone X) | 1 |
| 4 | ✓ **Usable** | Kīlauea Neighborhood Center | 2460 Keneke Street, Kīlauea | Outside FEMA high-risk flood zones (zone X) | 1 |
| 5 | ✓ **Usable** | Waimea Canyon Middle School | 9555 Huakai Road, Waimea | Outside FEMA high-risk flood zones (zone X) | 1 |

Source: county announcements of sites opened in 2026; FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM); tsunami evacuation zones (Hawaiʻi Statewide GIS).

Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

Candidate public schools in this county usable for this hazard: **16**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

## Care facilities in hazard zones

Care facilities in hazard zones – Kauaʻi, Coastal flooding (6 of 36)

| Name | Type | Zone |
|---|---|---|
| Aloha School Early Learning Center | Childcare centre | FEMA flood zone AE |
| Ke Kula Niʻihau O Kekaha PCS | Childcare centre | FEMA flood zone AH |
| Kekaha Elementary School | Childcare centre | FEMA flood zone AH |
| Kekaha Head Start | Childcare centre | FEMA flood zone AE |
| St Theresa Early Learning Academy | Childcare centre | FEMA flood zone AE |
| The Koloa Early School | Childcare centre | FEMA flood zone AE |

Source: CMS Provider Data Catalog (dialysis facilities, nursing homes, hospitals; facility addresses only), located with Esri ArcGIS World Geocoder; zones: State tsunami evacuation zone and FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM).

Source: Ready Keiki Locator, all pre-K sites (Office of the Lieutenant Governor, published through Hawaiʻi Statewide GIS): licensed group child care, public pre-K and charter pre-K sites only – no family child care homes; zones: State tsunami evacuation zone and FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM).

## 5. Needs and proposed partners

Likely estimate. Roles are **proposed – to be confirmed with partner**. The pledged column stays blank until a partner confirms it.

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Cots / sleeping space | 960 | cots<br>1 per person in shelter (planning assumption; no published per-person cot factor cited) | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Floor space (≤72 h) | 19,186 | sq ft<br>Florida DEM (2010): 20 sq ft per person | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Floor space (≤2 weeks) | 38,372 | sq ft<br>Florida DEM (2010): 40 sq ft per person | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Water | 960 | gal/day<br>Florida DEM (2010): 1 gal per person per day | American Red Cross | Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community | — | confirm with partner |
| Meals | 1,919 | meals/day<br>Florida DEM (2010): 2 meals per person per day | American Red Cross | Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Toilets (≤72 h) | 24 | toilets<br>Florida DEM (2010): 1 per 40 persons | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Toilets (≤2 weeks) | 48 | toilets<br>Florida DEM (2010): 1 per 20 persons | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Shelter staff | 68 | staff per shift<br>Florida DEM (2010) staffing ratios per shift: registration 1:165, dormitory 1:100, meals 1:110, mental health 1:250, health 1:200, security 1:100, sanitation 1:200, each at least 1; children's area 1:12 | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Office of Hawaiian Affairs (OHA), The Salvation Army, Philanthropy (foundations and donors), Business community | — | confirm with partner |
| Power outlets for power-dependent residents | 5 | outlets<br>1 per power-dependent person in shelter (planning assumption; HHS emPOWER-based estimate) | Hawai‘i Department of Health (DOH) | County Emergency Management, Health care organizations (incl. managed health care), Utilities | — | confirm with partner |
| Accessible transport | 852 | people in no-vehicle households at risk<br>ACS via CDC SVI; outreach upper bound, not all will need a ride | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), Business community, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Pet space – dog households | 136 | households with a dog AVMA (2020 data): 44.6% of households | County Emergency Management | Hawai‘i VOAD (Voluntary Organizations Active in Disaster) | — | confirm with partner |
| Pet space – cat households | 79 | households with a cat AVMA (2020 data): 26% of households | County Emergency Management | Hawai‘i VOAD (Voluntary Organizations Active in Disaster) | — | confirm with partner |
| Children's area | 226 | children in shelter tract share age 17 and under; staff counted in Shelter staff | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i Department of Health (DOH), Health care organizations (incl. managed health care), Philanthropy (foundations and donors) | — | confirm with partner |
| Interpreters – Ilocano | 10 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Kaua‘i + Maui); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Interpreters – Tagalog/Filipino | 6 | people with limited English in shelter<br>CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Kauaʻi + Maui); interpreter ratio not sourced – partner to set | Hawaiʻi Department of Human Services (DHS) | County Emergency Management, Hawaiʻi Emergency Management Agency (HI-EMA), Hawaiʻi State Public Library System, Hawaiʻi Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Spanish | 2 | people with limited English in shelter<br>CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Kauaʻi + Maui); interpreter ratio not sourced – partner to set | Hawaiʻi Department of Human Services (DHS) | County Emergency Management, Hawaiʻi Emergency Management Agency (HI-EMA), Hawaiʻi State Public Library System, Hawaiʻi Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Japanese | 1 | people with limited English in shelter<br>CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Kauaʻi + Maui); interpreter ratio not sourced – partner to set | Hawaiʻi Department of Human Services (DHS) | County Emergency Management, Hawaiʻi Emergency Management Agency (HI-EMA), Hawaiʻi State Public Library System, Hawaiʻi Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Interpreters – Marshallese | 1 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Kauaʻi + Maui); interpreter ratio not sourced – partner to set | Hawaiʻi Department of Human Services (DHS) | County Emergency Management, Hawaiʻi Emergency Management Agency (HI-EMA), Hawaiʻi State Public Library System, Hawaiʻi Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Hawaiian | 1 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Kauaʻi + Maui); interpreter ratio not sourced – partner to set | Hawaiʻi Department of Human Services (DHS) | County Emergency Management, Hawaiʻi Emergency Management Agency (HI-EMA), Hawaiʻi State Public Library System, Hawaiʻi Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

Source: likely estimate × planning factors (section 8; Florida Division of Emergency Management 2010; AVMA (2020 data) for pets).

**Service animals stay with their handler (ADA); not counted as pets.**

## Also plan for (county counts, not included in the totals above)

| Group | People | Detail | Date |
|---|---|---|---|
| People experiencing homelessness (Point-in-Time count) | 516 | sheltered 76, unsheltered 440; one-night count | 2026-01-25 |
| Visitors on island (average per day) | 28,577 | Kauaʻi | 2025 (annual average) |
| People in dam evacuation zones | 8,369 | State dam evacuation zones | zones fetched 2026-10-28 |

Source: Point-in-Time counts – Bridging the Gap 2026 report (Hawaiʻi, Maui, Kauaʻi; night of Jan 25, 2026) and Partners In Care 2026 count for Oʻahu as released by the City and County of Honolulu; visitors – DBEDT State of Hawaiʻi Data Book 2025, Table 7.06. Details: section 8.

State dam evacuation zones show the area that could flood if a dam fails; they are planning zones, not a forecast.

Source: State of Hawai‘i dam evacuation zones (published through Hawai‘i Statewide GIS). People = share of each Census area inside a zone × its population, the same method as for the tsunami zone. Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.

**Who else to plan for**

- **Domestic-violence survivors:** Served through confidential providers; no counts by design.
- **Off-grid households:** No public Census-area data; county and health partners to supply.
- **People on dialysis or receiving home care:** No public Census-area data; county and health partners to supply.
- **Earthquake:** Not modeled; listed as a limitation in the plan.
- **Caregivers and personal care attendants:** Stay with the person they support; count them for cots, meals and space.
- **Behavioral health:** Mental-health staff at each shelter (ratio: section 8).
- **Ni‘ihau:** Privately owned island, no public shelter; plan with the landowner and Kaua‘i County.
- **Families afraid to seek shelter (including immigrant families):** The county publishes who may use shelters; trusted community messengers share it.

## 6. Decision points

Proposed – supports, does not replace, county plans.

| When | Trigger | Action | Owner (proposed) | Support |
|---|---|---|---|---|
| **BEFORE** | Annual pre-positioning review (no official product)<br>County (sets the date) | Mark coastal shelters in V/A zones for site check; choose inland alternates. | County Emergency Management (each county) | Hawai‘i Department of Education (DOE) – schools, Hawai‘i Department of Human Services (DHS) |
| **BEFORE** | Coastal Flood Warning<br>NOAA NWS Honolulu | Alert priority coastal tracts; confirm inland shelters and transport. | County Emergency Management (each county) | Hawai‘i Department of Transportation (DOT), Hawai‘i Department of Human Services (DHS) |
| **BEFORE** | Before each season: who may use shelters (no official product)<br>County (sets the date) | County confirms and publishes who may use shelters and what ID is needed; trusted community messengers share it. | County Emergency Management (each county) | Hawai‘i Department of Human Services (DHS), Faith-based organizations, Office of Hawaiian Affairs (OHA), Hawai‘i VOAD (Voluntary Organizations Active in Disaster) |
| **DURING** | Coastal Flood Warning in effect<br>NOAA NWS Honolulu | Use inland sites only; close sites in flood zones; move residents who need transport before high tide. | County Emergency Management (each county) | American Red Cross, Hawai‘i Department of Human Services (DHS), Hawai‘i Department of Transportation (DOT) |
| **AFTER** | Warning cancelled or all clear<br>County / NWS / PTWC | Recovery sheltering and feeding for displaced households. | American Red Cross | Hawai‘i VOAD (Voluntary Organizations Active in Disaster), The Salvation Army |

Source: National Weather Service, Central Pacific Hurricane Center and Pacific Tsunami Warning Center product definitions (links: section 8).

# 7. Priority outreach and languages

## Top 10 Census areas by number of people with needs

| # | Community (Census area) | People in the risk area | Use power-dependent medical equipment (estimate) | People with a disability | Speak English less than well | People in households without a car | Needs total (a person can be in more than one column) |
|---|---|---|---|---|---|---|---|
| 1 | Līhuʻe<br>Census area 405 | 7,301 | 33 | 856 | 276 | 120 | 1,285 |
| 2 | Kekaha / Waimea<br>Census area 409 | 5,172 | 26 | 410 | 140 | 365 | 940 |
| 3 | Kōloa / Poʻipū<br>Census area 406.03 | 2,972 | 19 | 284 | 49 | 254 | 607 |
| 4 | Hāʻena / Hanalei<br>Census area 401.04 | 1,278 | 7 | 190 | 0 | 113 | 310 |
| 5 | Kapaʻa<br>Census area 403.02 | 2,463 | 12 | 175 | 22 | 0 | 209 |

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

## More needs in the largest communities

| Community (Census area) | Hearing difficulty | Vision difficulty | Cognitive difficulty (memory, decisions) | Age 65 and older, living alone |
|---|---|---|---|---|
| Līhuʻe<br>Census area 405 | 297 | 228 | 326 | 312 |
| Kekaha / Waimea<br>Census area 409 | 81 | 65 | 135 | 247 |
| Kōloa / Poʻipū<br>Census area 406.03 | 91 | 27 | 70 | 135 |

Source: ACS 2020–2024 5-year Census-area tables B18102 (hearing), B18103 (vision), B18104 (cognitive) and B09020 (age 65 and older living alone), times the share of each area in the risk area; homes in 10+ unit buildings: CDC/ATSDR SVI 2022 (E_MUNIT, housing units).

## How to reach residents

| Channel | Owner (proposed) | Support (proposed) |
|---|---|---|
| Official alerts: Wireless Emergency Alerts / Emergency Alert System and county alert sign-ups | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA) |
| Broadcast radio, including in-language programs where available | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA) |
| Outdoor warning sirens: a signal to turn on radio, TV or phone alerts for instructions | Hawai‘i Emergency Management Agency (HI-EMA) | County Emergency Management |
| Trusted community messengers: faith-based groups, Native Hawaiian organizations, health care providers, community nonprofits | Hawai‘i Department of Human Services (DHS) | Faith-based organizations, Office of Hawaiian Affairs (OHA), Health care organizations (incl. managed health care), Hawai‘i VOAD (Voluntary Organizations Active in Disaster) |
| Printed notices at shelters, libraries and schools, in the county's top languages | County Emergency Management (each county) | Hawai‘i State Public Library System, Hawai‘i Department of Education (DOE) – schools |

Source: proposed channels and formats in the plan's communication pathways – supports, does not replace, county and State public-information plans.

Populated Census areas whose land point is farther than 3,400 ft from every outdoor siren (HI-EMA manufacturer rating; actual range varies): **14**.

**Populated Census areas with less than 50% of people within rated siren range (4): prioritise WEA, radio and trusted messengers.** Kapa‘a (403.01); Lāwa‘i / ‘Ōma‘o (406.04); Princeville (401.06); Wailua Homesteads / Kapa‘a (402.04)

Source: Hawai‘i Emergency Management Agency public siren map (locations of permanent outdoor warning sirens; no live status). Rated range: 3,400 feet, the manufacturer rating published by the Hawai‘i Emergency Management Agency; the real range varies with terrain, buildings and weather. In range: within 3,400 feet of a siren, the same method as for the tsunami zone. Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.

### Accessible formats for every message:

- **ASL video:** short American Sign Language video of each message
- **Captions:** captions on every video and broadcast
- **Large print:** printed notices in large print
- **Plain language:** short sentences, one action at a time
- **Screen-reader-friendly text:** plain text or tagged web pages, not images of text

## Languages: people age 5 and older who speak English less than "very well" (Kaua‘i + Maui)

Kauaʻi and Maui share one language list: the Census survey reports both counties (and Kalawao) as one area, so they cannot be separated.

| Language | People | Margin of error (90%) |
|---|---|---|
| Ilocano | 9,131 | ±1,439 |
| Tagalog/Filipino | 5,074 | ±926 |
| Spanish | 1,711 | ±554 |
| Japanese | 1,339 | ±417 |
| Marshallese | 633 | ±488 |
| Hawaiian | 515 | ±289 |

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawaiʻi Data Book 2024, Table 1.44).

Draft messages are available statewide in Chinese, Hawaiian, Ilocano, Japanese, Korean, Pidgin, Spanish, Tagalog and Vietnamese; Chuukese and Marshallese need translators.

## All languages

All languages in the survey for Kauaʻi + Maui: people age 5 and older who speak English less than "very well"

| Language | People | Margin of error (90%) | Reliability |
|---|---|---|---|
| Ilocano | 9,131 | ±1,439 | |
| Tagalog/Filipino | 5,074 | ±926 | |
| Spanish | 1,711 | ±554 | |
| Japanese | 1,339 | ±417 | |
| Marshallese | 633 | ±488 | low reliability |
| Hawaiian | 515 | ±289 | |
| Chinese | 510 | ±266 | |
| Korean | 360 | ±210 | |
| Vietnamese | 359 | ±216 | |
| Tongan | 320 | ±212 | low reliability |
| Other Eastern Malayo-Polynesian languages | 277 | ±178 | |
| Chuukese | too few to estimate | | low reliability |
| Cebuano and other Philippine languages | 259 | ±192 | low reliability |
| Portuguese | too few to estimate | | low reliability |
| Pidgin (Hawaiʻi Creole English) | 220 | ±192 | low reliability |
| German | too few to estimate | | low reliability |
| Thai | too few to estimate | | low reliability |
| Chamorro | too few to estimate | | low reliability |
| Sinhala | too few to estimate | | low reliability |
| Greek | too few to estimate | | low reliability |
| Russian | too few to estimate | | low reliability |
| Dutch | too few to estimate | | low reliability |
| Khmer | too few to estimate | | low reliability |
| Samoan | too few to estimate | | low reliability |
| French | too few to estimate | | low reliability |
| Hungarian | too few to estimate | | low reliability |

| Language | People | Margin of error (90%) | Reliability |
|---|---|---|---|
| Italian | too few to estimate | | low reliability |
| Latvian | too few to estimate | | low reliability |
| Afrikaans | too few to estimate | | low reliability |
| Akan (including Twi) | too few to estimate | | low reliability |
| Arabic | too few to estimate | | low reliability |
| Bulgarian | too few to estimate | | low reliability |
| Danish | too few to estimate | | low reliability |
| Farsi | too few to estimate | | low reliability |
| Finnish | too few to estimate | | low reliability |
| Hebrew | too few to estimate | | low reliability |
| Hindi | too few to estimate | | low reliability |
| Hmong | too few to estimate | | low reliability |
| Indonesian | too few to estimate | | low reliability |
| Lao | too few to estimate | | low reliability |
| Malay | too few to estimate | | low reliability |
| Mongolian | too few to estimate | | low reliability |
| Norwegian | too few to estimate | | low reliability |
| Other Indo-European languages | too few to estimate | | low reliability |
| Other and unspecified languages | too few to estimate | | low reliability |
| Polish | too few to estimate | | low reliability |
| Punjabi | too few to estimate | | low reliability |
| Serbian | too few to estimate | | low reliability |
| Swedish | too few to estimate | | low reliability |
| Swiss German | too few to estimate | | low reliability |
| Tamil | too few to estimate | | low reliability |
| Turkish | too few to estimate | | low reliability |
| Yiddish | too few to estimate | | low reliability |

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawai‘i Data Book 2024, Table 1.44).

Low reliability: margin of error large compared with the estimate (coefficient of variation above 40%), or too few to estimate (fewer than 100 people or 10 survey responses).

Pre-approved message templates: none yet; no machine translation without human review.

# 8. Sources and assumptions

## Planning factors

| Factor | Value | Source or team assumption |
|---|---|---|
| Shelter-seeking rate – Hurricane | 2% / 3% / 5% | South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%) |
| Shelter-seeking rate – Wildfire | 2% / 5% / 11.2% | high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are Team assumption. |
| Shelter-seeking rate – Tsunami | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Inland flooding | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Coastal flooding | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Footprint – Hurricane | Every Census area | Hurricanes affect whole islands (Lala, Lowell 2026). |
| Footprint – Tsunami | Share living inside the tsunami evacuation zone | Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately. |
| Footprint – Wildfire | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Footprint – Inland flooding | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Footprint – Coastal flooding | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Floor sqft 72h | 20 | Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h |
| Floor sqft 2wk | 40 | Florida DEM (2010): short-term shelter, <=2 weeks |
| Water gal per day | 1 | Florida DEM (2010) |
| Meals per day | 2 | Florida DEM (2010): two hot meals a day plus snacks |

| Factor | Value | Source or team assumption |
|---|---|---|
| Persons per toilet short term | 20 | Florida DEM (2010): short-term shelter, <=2 weeks |
| Power outlets per power dependent person | 1 | Team assumption |
| Persons per toilet 72h | 40 | Florida DEM (2010): emergency evacuation shelter, <=72 h |
| Staffing ratios (one staff member per N residents) | client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12 | One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations. |
| Dog owning household share | 0.446 | AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27. |
| Cat owning household share | 0.26 | AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'. |

## Data sources

| Source | Fetched | URL / file |
|---|---|---|
| CDC/ATSDR Social Vulnerability Index 2022, census tracts | 2026-10-27 | https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2 |
| FEMA National Risk Index v1.20, census tracts – Hurricane | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| FEMA National Risk Index v1.20, census tracts – Wildfire | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| FEMA National Risk Index v1.20, census tracts – Tsunami | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| FEMA National Risk Index v1.20, census tracts – Inland Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |

| Source | Fetched | URL / file |
|---|---|---|
| FEMA National Risk Index v1.20, census tracts – Coastal Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2 |
| Extreme Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12 |
| HHS emPOWER – electricity-dependent DME, county level | 2026-10-27 | https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2 |
| ACS 2020-2024 5-year, tract tables B18102 (hearing), B18103 (vision), B18104 (cognitive), B09020 (age 65+ living alone) | 2026-10-26 | https://data.census.gov/api/access/data/table?g=040XX00US15%241400000&id=ACSDT5Y2024.B18102 |
| Share of people in a zone (tsunami, dam evacuation zones, siren range) | | Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population. |
| Hawai‘i Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26) | 2026-10-28 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6 |
| County shelter activations 2026 (official county releases) | 2026-10-27 | County shelter activation table: one official county source per row |
| 2020 Census Block Assignment File, place/CDP (tract community labels) | 2026-10-27 | https://www2.census.gov/geo/docs/maps-data/data/baf2020/BlockAssign_ST15_HI.zip |
| 2020 Census PL 94-171 block population P1_001N (tract community labels) | 2026-10-27 | https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Hawaii/hi2020.pl.zip |
| ACS PUMS 2019-2023 5-year (detailed languages) | 2026-10-27 | https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip |
| DBEDT State of Hawai‘i Data Book 2024, Table 1.44 (validation) | 2026-10-27 | https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf |
| Partner list: Hawai‘i DHS – ESF-6 challenge statement (2026) | 2026-10-15 | Hawai‘i DHS – ESF-6 challenge statement (2026) |
| pre season hurricane | | https://www.nhc.noaa.gov/cphcmission.php |
| hurricane watch | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| hurricane warning | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| tsunami warning | | https://www.tsunami.gov/?page=message_definitions |
| flash flood warning | | https://www.weather.gov/safety/flood-watch-warning |
| red flag warning | | https://www.weather.gov/safety/wildfire-ww |

| Source | Fetched | URL / file |
|---|---|---|
| coastal flood warning | | https://www.weather.gov/okx/marinedef |
| coastal flood warning in effect | | https://www.weather.gov/okx/marinedef |
| Ready Keiki Locator – All Pre-K Sites (Office of the Lieutenant Governor, hosted by Hawai‘i Statewide GIS) | 2026-10-28 | https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/Ready_Keiki_GIS/FeatureServer/0 |
| Hawai‘i Emergency Management Agency – all-hazard outdoor warning sirens (public siren layer; locations only) | 2026-10-28 | https://services9.arcgis.com/aKxrz4vDVjfUwBWJ/arcgis/rest/services/public_siren_status_layer_view/FeatureServer/0 |
| State of Hawai‘i dam evacuation zones (hosted by Hawai‘i Statewide GIS) | 2026-10-28 | https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/DamEvacZones_joined/FeatureServer/0 |
| Hawai‘i Emergency Management Agency – siren audible radius (3,400 feet, manufacturer rating) | 2026-10-28 | https://dod.hawaii.gov/hiema/all-hazard-statewide-outdoor-warning-siren-system/ |
| Tsunami Safe Zones (Hawai‘i Statewide GIS, Hazards layer 13) (Hawai‘i Island March 2025 (corrected November 2025), O‘ahu April 2025, Kaua‘i March 2025, Maui, Moloka‘i and Lāna‘i 2013) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/13 |
| Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawai‘i Statewide GIS, Hazards layer 7) (2007; boundaries confirmed by DLNR-DOFAW 2017 and 2022) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/7 |
| Sea Level Rise – Potential Flooded Highways, 3.2 ft scenario (Hawai‘i Sea Level Rise Viewer, PacIOOS, via Hawai‘i Statewide GIS, Climate layer 71) (Hawai‘i Sea Level Rise Vulnerability and Adaptation Report, 2017) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Climate/MapServer/71 |

| Source | Fetched | URL / file |
|---|---|---|
| Pacific Disaster Center – Hawaii Emergency Shelters (capacity: American Red Cross Hawai‘i State Chapter) (ground condition 2009-10-08 (PDC metadata)) – via third-party ArcGIS Online copy; PDC catalogue confirms 2009 date | 2026-10-30 | https://services1.arcgis.com/x4h61KaW16vFs7PM/arcgis/rest/services/Potential_Emergency_Shelters_PDC/FeatureServer/36 |
| Hawai‘i DOT Highway Performance Monitoring System road centrelines (Hawai‘i Statewide GIS, Transportation layer 12) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Transportation/MapServer/12 |
| Hawai‘i DOT surveyed mile markers (road-closure point, Route 11 in Ka‘ū) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Transportation/MapServer/14 |
| DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census) | 2026-10-27 | https://files.hawaii.gov/dbedt/economic/databook/db2025/DB2025_section07.xlsx |
| CMS Provider Data Catalog – Dialysis Facility – Listing by Facility | | https://data.cms.gov/provider-data/dataset/23ew-n7w9 |
| CMS Provider Data Catalog – Nursing Homes – Provider Information | | https://data.cms.gov/provider-data/dataset/4pq5-n9py |
| CMS Provider Data Catalog – Hospital General Information | | https://data.cms.gov/provider-data/dataset/xubh-q36u |
| Esri ArcGIS World Geocoder (site check, care facilities) and ArcGIS routing (drive and walk times, normal traffic). Powered by Esri. | | https://developers.arcgis.com/rest/ |

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.

## 9. Results

- **Coverage:** 436 of 436 Census areas have needs data (FEMA risk data joined for 435 of 436). Source: coverage check of the needs layer (CDC SVI 2022, FEMA NRI v1.20).
- **Backtest – Lahaina wildfire (2023-08-11):** observed 1,418 people in shelters; model 254 / 635 / 1,423 (low / likely / high). This is a calibration event, not validation: the high rate was set from it. Source: Lahaina 2023 backtest (Maui Now count of people in shelters).

# 10. Stress tests – summary

**Stress test, not a forecast.** Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results.

People counted: residents of Census areas whose drive from the area's centre point to the nearest usable 2026 site was 15 minutes or less before the loss and is more than 15 minutes (or no route) after it; areas already beyond 15 minutes are not counted.

Top 3 for Kauaʻi – Coastal flooding and all-hazard tests, ranked by people affected (at most two from one test). Full results: the **Kauaʻi stress-test annex** (a separate PDF, also linked in the app).

1. **Stranded visitors for 3 or 7 days – Kauaʻi**
   28,577 visitors on an average day (island level).
2. **All four counties at once (coastal flooding)**
   17,661 people who may need a public shelter (likely), counties together (cots: 17,664; shelter staff per shift: 1,171).
3. **Kauaʻi and Honolulu at once (coastal flooding)**
   13,129 people who may need a public shelter (likely), counties together (cots: 13,131; shelter staff per shift: 868).

Sources, evidence and team assumptions for each result: in the annex.

## Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?
3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?